1.

Item	Factor of Production	Explanation
A. A teacher working in a school	**Labour**	Human effort used in production
B. A sewing machine in a textile factory	**Capital**	Man-made tool used in production
C. A plot of land used for farming	**Land**	Natural resource
D. An investor starting a new tech startup	**Entrepreneurship**	Organizer and risk-taker in production
E. A river used for hydroelectric power generation	**Land**	Natural resource

2.

Decision	Opportunity Cost	Explanation
A. Watching a movie instead of studying	**The study time or potential grade improvement**	Gave up study time
B. Baking cakes instead of bread	**The bread that could have been baked**	Alternative forgone
C. Investing in stocks instead of fixed deposit	**Interest income from fixed deposit**	Missed return
D. Buying a smartphone instead of laptop	**The laptop**	Foregone choice
E. Taking unpaid leave to go on vacation	**Lost income**	Wages forgone

4.

Country	Wheat	Cloth
A	10	20
B	5	10

(A) Absolute Advantage:

- **Country A** has an absolute advantage in both goods (produces more of both with same resources).

(B) Opportunity Cost:

Country	1 Wheat costs	1 Cloth costs
A	2 Cloth / 10 → **2 Cloth**	½ Wheat
B	10/5 = **2 Cloth** /? Wait — let's compute properly	½ Wheat

Let's recompute carefully:

- For A:
  - 1 Wheat = 2 Cloth / 10 = 2 Cloth → correct.
  - 1 Cloth = ½ Wheat.

- For B:
    - 1 Wheat = (10/5) = **2 Cloth**.
    - 1 Cloth = ½ Wheat.

So they are identical — **no comparative advantage** (opportunity cost same).

5.

Price	**Qd**	**Qs**
10	50	10
20	40	20
30	30	30
40	20	40
50	10	50

(A) Graph:

- Downward sloping demand, upward sloping supply.
- Equilibrium at **P = 30**, **Q = 30**.

(B) Labels:

- **Equilibrium Price:** 30
- **Equilibrium Quantity:** 30
- **Surplus:** Above equilibrium (Qs > Qd)
- **Shortage:** Below equilibrium (Qd > Qs)
- **Consumer Surplus:** Area above P_e and below demand curve
- **Producer Surplus:** Area below P_e and above supply curve

6.

dogs (normal) & Butter Buns (inferior)

- Income ↓ (due to tax ↑)
- Demand for **normal goods (hotdogs)** ↓ → **Price and quantity fall**.
- Demand for **inferior goods (butter buns)** ↑ → **Price and quantity rise**.

7.

Situation:

- Fuel prices ↑ → demand for electric bicycles ↑ (rightward shift in demand)
- Raw material cost ↑ → supply of electric bicycles ↓ (leftward shift in supply)

Diagram:

- Demand curve shifts **rightward**, supply curve shifts **leftward**.

Effect:

- Price: **Increases**
- Quantity: **Uncertain** (depends on magnitude of shifts)

8.

Product	**June (P,Q)**	**Sept (P,Q)**
A	40, 500	45, 550
B	55, 800	60, 780

(A) Price Elasticity of Demand (PED)

$$PED = \frac{\%\Delta Q}{\%\Delta P} = \frac{(550 - 500)/((550 + 500)/2)}{(45 - 40)/((45 + 40)/2)} = \frac{50/525}{5/42.5} = \frac{0.0952}{0.1176} = 0.81$$

PED = 0.81 (Inelastic)
→ Demand not very responsive to price.

(B) Cross Price Elasticity of Demand (XED)

Price of B ↑ from 55 → 60
Quantity of A ↑ from 500 → 550

$$XED = \frac{\%\Delta Q_A}{\%\Delta P_B} = \frac{(550 - 500)/525}{(60 - 55)/57.5} = \frac{0.0952}{0.0870} = 1.09$$

XED = +1.09 (Positive, substitutes)
→ Product A and B are **substitute goods**.